

The simple average return on small stocks is more than 4 percent higher than its compounded return. The reason is small stocks' higher price volatility. Greater variation in returns reduces long-term compounded returns exponentially. The simple average return on T-bills is about that same as the compounded return because T-bills had a very small standard deviation of returns.

Volatility creates lower returns and thus is itself a risk. If you can reduce the volatility in a portfolio, then the compounded return moves higher, closer to the simple average return of the weighted investments in the portfolio. This is how lower portfolio price volatility increases portfolio return over time. We'll go through some math to get to a better understanding of this important concept.

Refer to [Table 2-2](#) for an illustration on how the variability of annual returns reduces the account value over time. Four portfolios are presented. Each portfolio starts with \$10,000.

An analysis of the four portfolios in [Table 2-2](#) confirms that all the accounts had a simple average annual return of 5.0 percent, yet each one had a different compounded return also known as the annualized return. Portfolio A's return had a 1.1 percent higher compounded return than Portfolio D because it had less annual volatility. The reason for the difference is *how* Portfolio A earned a 5.0 percent simple average relative to Portfolio D. Portfolio A had very consistent annual returns (+5 percent, +5 percent). Portfolio D had high variability in annual returns (+20 percent, -10 percent). Consequently, while the simple averages of Portfolio A and Portfolio D were the same (+5 percent), Portfolio A had the highest ending value, and Portfolio D compounded to the lowest.

The standard measure of volatility is the standard deviation. Think of standard deviation as the "average miss" from the portfolio's simple average return. A portfolio can have a simple average return of 5 percent even though it never actually earns 5 percent in any given year. The average annual miss from 5 percent is the portfolio's annual standard deviation.

TABLE 2-2

Four Portfolios with Different Standard Deviations